Robust and Cross-Dataset Evaluation for Machine Learning-Based Intrusion Detection: Statistical Validation, Modern Models, Moderns Datasets, and Multi-Criteria Ranking

Engineering / Master's Project

École Supérieure en Informatique – Sidi Bel Abbès (ESI-SBA)

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Research Domain: Cybersecurity – Artificial Intelligence – Intrusion Detection Systems (IDS) – Network Intrusion Detection – Machine Learning – Multiclass Classification – Binary Classification – NSL-KDD – UNSW-NB15 – CICIDS2017 – Feature Selection – Class-Imbalance Handling.

1 Project Context

Machine learning (ML) has extensively investigated for intrusion detection systems (IDS). Numerous studies have evaluated classical and modern ML models using benchmark network intrusion datasets. However, reported performance can depend strongly on several experimental choices, including the classifier, feature-selection method, class-imbalance strategy, classification granularity, evaluation metrics, and ranking procedure.

A previous study developed a systematic experimental framework for evaluating ML-based intrusion detection using the NSL-KDD dataset. The framework considered binary and multiclass classification, several feature-selection methods, multiple class-imbalance strategies, nine machine learning models, and a multi-metric evaluation protocol.

The present project proposes an extension of this framework toward a more robust empirical evaluation. The project will investigate the stability of the experimental conclusions across repeated executions, datasets, modern ML models, feature-selection strategies, imbalance-handling methods, and alternative multi-criteria ranking schemes.

The project will therefore move from a benchmark-centered evaluation toward a broader and statistically supported experimental study.

2 Scientific Positioning

The project does not aim to claim a fundamentally new intrusion detection algorithm. Instead, its main contribution will be positioned as:

> *A systematic factorial evaluation of the interaction among classifier, feature-selection method, imbalance strategy, and classification granularity under a unified multi-metric evaluation protocol.*

The proposed extension will investigate the robustness and reproducibility of the resulting conclusions across datasets, random seeds, modern learning algorithms, and ranking assumptions.

3 Problem Statement

The existing experimental framework provides a broad comparison of machine learning configurations. However, several questions remain open:

1. Are the observed performance differences statistically significant?
2. Do the best configurations remain competitive across independent executions?
3. Are the conclusions obtained from NSL-KDD preserved on more recent intrusion-detection datasets?
4. Do modern models such as XGBoost, LightGBM, CatBoost, MLP, CNN, LSTM, or Transformer-based models outperform the classical approaches consistently?
5. How sensitive is the final configuration ranking to the selected metric weights?
6. How sensitive is the ranking to the normalization procedure?
7. Which attack categories remain difficult to detect, particularly minority classes such as R2L and U2R?
8. Which interactions among classifier, feature selection, imbalance strategy, and classification granularity explain the observed performance?
9. Can the complete experimental process be automated and reproduced?

4 General Objective

The main objective is to extend the existing intrusion-detection evaluation framework into a robust, reproducible, and cross-dataset experimental platform for machine learning-based IDS.

The project will combine:

- statistical validation;
- cross-dataset evaluation;
- modern machine learning models;
- feature-selection strategies;
- class-imbalance strategies;
- class-level performance analysis;
- sensitivity analysis for multi-criteria ranking;
- ranking-stability analysis;
- reproducible experimental execution.

5 Specific Objectives

The project will pursue the following objectives.

5.1 Objective 1: Statistical Validation

The experiments will be repeated using multiple independent random seeds. At least ten independent executions will be considered for important configurations.

For each configuration and metric, the project will report:

$$\text{Mean} \pm \text{Standard Deviation.}$$

Where appropriate, statistical tests will be applied to determine whether observed differences between configurations are significant.

Possible statistical procedures include:

- paired statistical tests;
- post-hoc pairwise comparisons;
- effect-size measures;
- confidence intervals.

The selection of statistical procedures will depend on the experimental design and distributional assumptions.

5.2 Objective 2: Cross-Dataset Evaluation

The project will extend the evaluation beyond NSL-KDD.

The minimum configuration will include:

- NSL-KDD;
- UNSW-NB15.

An extended configuration may additionally include:

- CICIDS2017;
- CSE-CIC-IDS2018.

The objective is not simply to compare datasets, but to determine whether conclusions concerning classifiers, feature selection, imbalance handling, and ranking remain consistent across different network intrusion benchmarks.

5.3 Objective 3: Evaluation for Modern ML Models

The existing classical models will be complemented with more recent machine learning approaches.

At minimum, the following models will be investigated:

- XGBoost;
- LightGBM;
- CatBoost;
- Multilayer Perceptron (MLP).

Depending on available computational resources, at least one deep learning architecture will also be investigated, such as:

- CNN;
- LSTM;
- Transformer-based model.

The objective will be to determine whether modern models provide consistent improvements rather than relying on a single benchmark result.

5.4 Objective 4: Feature Selection Evaluation

The existing feature-selection strategies will be retained to preserve comparability with the previous study.

The project may consider:

- variance-based selection;
- chi-square selection;
- ANOVA;
- mutual information;
- Random Forest-based selection;
- Extra Trees-based selection;
- Logistic L1-based selection.

Additional methods may be investigated if computational resources permit.

The analysis will focus not only on the final performance but also on the interaction between feature-selection strategy and classifier.

5.5 Objective 5: Class-Imbalance Evaluation

The project will retain the previously investigated imbalance strategies and may introduce additional approaches where appropriate.

The evaluation will consider strategies such as:

- Random Over-Sampling;
- Random Under-Sampling;
- SMOTE;
- Borderline-SMOTE;
- SVM-SMOTE;
- ADASYN;
- SMOTE-Tomek;
- SMOTE-ENN;
- ENN;
- NearMiss.

The objective is to determine which strategies provide stable improvements and for which classifiers and datasets.

6 Objective 6: Class-Level Multiclass Analysis

Particular attention will be given to multiclass intrusion detection.

A relatively high overall F1-score does not necessarily indicate homogeneous performance across attack categories. Therefore, the project will provide:

- precision per class;
- recall per class;
- F1-score per class;
- support per class;
- confusion matrices;
- macro-F1;
- weighted-F1;
- balanced accuracy;
- class-specific error analysis.

Special attention will be given to minority attack categories, particularly R2L and U2R in NSL-KDD.

The analysis will investigate whether high aggregate performance is associated with reduced recognition of minority classes.

7 Objective 7: Multi-Criteria Ranking Sensitivity

The previous framework uses a weighted multi-metric score to rank configurations. However, predefined metric weights may influence the final ranking.

The project will therefore investigate several weighting scenarios.

Table 1: Proposed multi-criteria ranking scenarios.

Scenario	Main principle
A	Equal metric weights
B	F1-oriented weighting
C	Recall-oriented weighting
D	Imbalance-oriented weighting
E	Original weighting scheme
F	Alternative expert-defined weighting

For each scenario, the configuration ranking will be recomputed.

The stability of the ranking will then be investigated using measures such as:

- rank correlation;
- top-k configuration overlap;
- rank displacement;
- frequency of appearance among the top-ranked configurations.

The goal is to determine whether the principal conclusions remain stable under reasonable modifications to metric weights.

8 Objective 8: Robustness for Metric Normalization

The current ranking procedure uses normalized metric values. Since min-max normalization depends on the minimum and maximum values observed within the experimental space, adding a new model or configuration may change normalized values and consequently alter the ranking.

The project will therefore investigate this issue explicitly.

Different experimental spaces will be considered:

- baseline models only;
- classical models with feature selection;
- classical models with class-imbalance strategies;
- complete experimental space;
- complete space including modern models.

The resulting rankings will be compared to determine whether the conclusions depend strongly on the selected experimental space.

9 Experimental Protocol

The extended experimental protocol will follow the general structure:

$$\text{Dataset} \rightarrow \text{Preprocessing} \rightarrow \text{Binary/Multiclass Classification} \rightarrow \text{Feature Selection} \rightarrow$$
$$\text{Class-Imbalance Strategy} \rightarrow \text{ML Model} \rightarrow \text{Repeated Execution} \rightarrow \text{Multi-Metric Evaluation} \rightarrow$$
$$\text{Statistical Validation} \rightarrow \text{Ranking} \rightarrow \text{Ranking Stability.}$$

A strict separation between training and test data will be maintained.

Feature selection, oversampling, undersampling, and model training procedures must be applied only within the training data. The test data will remain untouched until the final evaluation stage.

This principle will be maintained for all datasets.

10 Experimental Factors

The experimental design will consider the following factors:

Table 2: Main experimental factors.

Factor	Investigated configurations
Dataset	NSL-KDD, UNSW-NB15, optional CICIDS2017/CSE-CIC-IDS2018
Classification	Binary, multiclass
Classifiers	Classical and modern ML models
Feature selection	Multiple filter and embedded methods
Imbalance handling	Multiple over-sampling and under-sampling strategies
Random seeds	At least 10 independent executions
Evaluation	Multi-metric evaluation
Ranking	Multiple weighting scenarios
Normalization	Multiple experimental spaces

11 Evaluation Metrics

The evaluation will include metrics appropriate for both balanced and imbalanced classification.

The main metrics will include:

- Accuracy;
- Balanced Accuracy;
- Precision;
- Recall;
- F1-score;
- Specificity;
- Kappa;
- ROC-AUC;
- PR-AUC;
- Log-Loss;
- Brier Score;

For multiclass experiments, macro-averaged and weighted metrics will be reported where appropriate.

12 Statistical Analysis

For each important experimental configuration, the project will execute repeated experiments using independent random seeds.

The resulting measurements will be summarized as:

$$\mu \pm \sigma,$$

where μ represents the mean and σ represents the standard deviation.

Statistical testing will be used to determine whether differences between selected configurations are supported by the repeated experiments.

The project will also report effect sizes where appropriate, allowing practical differences to be distinguished from statistically significant but small differences.

13 Interaction Analysis

A major extension will consist of studying interactions among experimental factors.

The analysis will investigate questions such as:

- Why does ANOVA perform particularly well with Gradient Boosting?
- Why can Random Forest-based feature selection benefit multiclass classification?
- Which classifiers benefit most from SVM-SMOTE?
- Under which conditions does feature selection reduce or improve performance?
- Are the best imbalance strategies classifier-dependent?

- Does classification granularity change the relative ranking of models?
- Are certain feature-selection strategies consistently associated with specific classifiers?

The objective is to move beyond reporting numerical results toward explaining observed patterns.

14 Cross-Dataset Generalization

For each selected experimental configuration, performance will be evaluated across datasets.

A comparison table will be developed with a structure such as:

Table 3: Cross-dataset comparison framework.

Configuration	Dataset	F1	Bal. Acc.	ROC-AUC	Rank	Stability
Configuration 1	NSL-KDD	–	–	–	–	–
Configuration 1	UNSW-NB15	–	–	–	–	–
Configuration 2	NSL-KDD	–	–	–	–	–
Configuration 2	UNSW-NB15	–	–	–	–	–

This analysis will identify configurations that perform consistently across datasets rather than configurations that perform well on only one benchmark.

15 Expected Scientific Contributions

The expected contributions of the project are:

1. An extended and reproducible experimental framework for ML-based intrusion detection.
2. Statistical validation of performance differences among experimental configurations.
3. A cross-dataset evaluation involving NSL-KDD and at least one additional intrusion-detection dataset.
4. An empirical comparison between classical and modern machine learning models.
5. A detailed class-level evaluation for multiclass intrusion detection.
6. A sensitivity analysis for multi-criteria metric weighting.
7. An investigation of ranking stability under alternative normalization settings.
8. An interaction analysis among classifiers, feature-selection methods, imbalance strategies, and classification granularity.
9. Identification of configurations that provide stable performance across datasets and experimental conditions.
10. A reproducible implementation capable of automatically executing and documenting the experimental pipeline.

16 Engineering Project Scope

The Engineering project will center around the implementation and integration aspects.

The engineering student will primarily focus on:

- designing the experimental software architecture;
- integrating multiple datasets;
- implementing the complete preprocessing pipeline;
- integrating classical and modern ML models;
- implementing feature-selection and imbalance strategies;
- implementing repeated experiments;
- automating experiment configuration;
- storing experimental results;
- generating tables and figures automatically;
- implementing the multi-criteria ranking module;
- implementing ranking sensitivity analysis;
- developing reproducible experiment logs;
- documenting the complete software framework.

The engineering deliverable should be a modular experimental platform that can easily add a new dataset, model, feature-selection method, imbalance strategy, metric, or ranking scenario.

17 Master's Project Scope

The Master's project will emphasize scientific investigation and interpretation.

The Master's student will focus on:

- designing the statistical experimental protocol;
- investigating the effect of random seeds;
- performing statistical significance testing;
- comparing classical and modern models;
- conducting cross-dataset experiments;
- studying class-level multiclass performance;
- investigating minority attack categories;
- conducting ranking sensitivity analysis;
- evaluating normalization sensitivity;
- studying interactions among experimental factors;
- interpreting the observed results;
- identifying configurations with stable performance;
- preparing a scientific manuscript based on the experimental findings.

18 Possible Engineering–Master's Coordination

The two projects can be coordinated around a common experimental platform.

Table 4: Complementary project responsibilities.

Engineering Project	Master's Project
Experimental platform	Experimental design
Dataset integration	Statistical validation
Pipeline automation	Cross-dataset evaluation
Model implementation	Scientific comparison
Feature-selection implementation	Class-level analysis
Imbalance strategy implementation	Ranking sensitivity
Result management	Ranking stability
Automatic table generation	Interpretation and discussion
Reproducibility tools	Scientific manuscript

19 Proposed Work Packages

WP1 – Literature Review

- Review recent ML-based IDS studies.
- Review NSL-KDD, UNSW-NB15, CICIDS2017, and CSE-CIC-IDS2018.
- Review feature-selection techniques.
- Review class-imbalance strategies.
- Review statistical validation procedures.
- Review multi-criteria decision and ranking methods.

WP2 – Experimental Framework Extension

- Refactor the existing experimental framework.
- Introduce a unified configuration system.
- Automate experiment execution.
- Implement reproducible random-seed management.
- Standardize result storage.

WP3 – Dataset Extension

- Integrate UNSW-NB15.
- Standardize preprocessing across datasets.
- Define binary and multiclass label mappings.
- Preserve strict training/test separation.
- Optionally integrate CICIDS2017 or CSE-CIC-IDS2018.

WP4 – Model Extension

- Integrate XGBoost.
- Integrate LightGBM.
- Integrate CatBoost.
- Integrate MLP.
- Optionally integrate CNN, LSTM, or Transformer models.

WP5 – Statistical Validation

- Repeat selected experiments using at least ten random seeds.
- Calculate means and standard deviations.
- Apply appropriate statistical tests.
- Calculate confidence intervals and effect sizes where appropriate.

WP6 – Class-Level Analysis

- Generate confusion matrices.
- Calculate per-class precision, recall, and F1-score.
- Investigate minority attack categories.
- Compare macro and weighted evaluation measures.

WP7 – Ranking Robustness and Scientific Evaluation

- Implement alternative metric-weighting schemes.
- Recompute configuration rankings.
- Compare ranking stability.
- Investigate normalization sensitivity.
- Identify consistently high-ranking configurations.
- Analyze interactions among experimental factors.
- Compare results across datasets.
- Identify stable configurations.
- Interpret the main findings.
- Prepare the final scientific report.

20 Expected Deliverables

The project will produce the following deliverables:

1. A modular and reproducible IDS experimental framework.
2. Source code for dataset preprocessing.
3. Source code for feature-selection methods.
4. Source code for class-imbalance strategies.
5. Implementations of classical and modern ML models.
6. Automated experiment configuration and execution.
7. Repeated-execution results.
8. Statistical analysis results.
9. Class-level multiclass evaluation.
10. Cross-dataset comparison tables.
11. Ranking sensitivity results.
12. Ranking-stability analysis.
13. Automatically generated figures and tables.
14. Engineering project report.
15. Master's dissertation/report.
16. Potential scientific publication.

21 Expected Results

The project is expected to answer the following questions:

- Which machine learning models provide stable performance across datasets?
- Which feature-selection methods are most consistently useful?
- Which class-imbalance strategies provide reliable improvements?
- Are performance differences statistically supported?
- Which configurations remain strong when the ranking weights change?
- Which configurations remain strong when additional models and datasets are introduced?
- Which attack categories remain difficult to detect?
- Does high aggregate F1 correspond to homogeneous multiclass performance?
- How strongly does the normalization strategy influence configuration ranking?

22 Potential Research Questions

The project can be formulated around the following research questions:

RQ1: How stable are ML-based IDS performance results across independent experimental executions?

RQ2: How do feature-selection and class-imbalance strategies interact with different classifiers?

RQ3: Do conclusions obtained from NSL-KDD remain valid on more recent intrusion-detection datasets?

RQ4: Do modern ML models provide consistent improvements over classical models across datasets and classification granularities?

RQ5: How sensitive is multi-criteria configuration ranking to alternative metric-weighting schemes?

RQ6: How sensitive is configuration ranking to the experimental space used for metric normalization?

RQ7: Which attack categories remain difficult to identify in multiclass intrusion detection?

RQ8: Which experimental configurations demonstrate the greatest cross-dataset and cross-seed stability?

23 Proposed Timeline

Table 5: Indicative project schedule.

Step	Main activities
Step 1	Literature review and analysis of existing framework
Step 2	Framework refactoring and dataset integration
Step 3	Feature-selection and imbalance experiments
Step 4	Integration of modern ML models
Step 5	Repeated experiments and statistical validation
Step 6	Cross-dataset experiments and class-level analysis
Step 7	Ranking sensitivity and stability analysis
Step 8	Scientific interpretation and comparison
Step 9	Report writing, final experiments, and defense preparation

24 Technical Environment

The project may use:

- Python;
- scikit-learn;
- imbalanced-learn;
- XGBoost;
- LightGBM;
- CatBoost;

- PyTorch or TensorFlow for deep learning experiments;
- pandas;
- NumPy;
- SciPy;
- Matplotlib;
- SHAP where explainability analysis is needed;
- Git for version control.

25 Ensuring Reproducibility

All important experiments must be reproducible.

The framework should record:

- dataset version;
- preprocessing configuration;
- feature-selection method;
- number of selected features;
- imbalance strategy;
- classifier and hyperparameters;
- random seed;
- evaluation metrics;
- execution time;
- software environment;
- final model performance;
- ranking configuration.

Each experimental configuration should be uniquely identifiable so that its results can be reproduced without manually reconstructing the experimental settings.

26 Limitations and Scientific Considerations

The project will explicitly recognize several limitations.

First, benchmark datasets cannot fully represent modern network traffic. Cross-dataset evaluation can reduce this limitation but cannot completely eliminate it.

Second, multi-criteria ranking remains dependent on the selected metrics, their weights, and the normalization procedure. The sensitivity analysis is therefore intended to quantify this dependency rather than assume a universally optimal ranking.

Third, computational constraints may limit the number of datasets, models, and repeated executions. The final experimental scope will therefore be defined according to available computational resources while preserving statistical validity.

Fourth, a higher aggregate score does not necessarily imply better detection for every attack category. Class-level results will therefore complement aggregate metrics.

27 Expected Final Contribution

The final project should provide a scientifically rigorous extension for the existing ML-based IDS evaluation framework.

Rather than proposing another isolated model comparison, the project will investigate the robustness of conclusions under changes in:

$$\boxed{\text{Dataset} \times \text{Classifier} \times \text{Feature Selection} \times \text{Imbalance Strategy} \times \text{Classification Granularity} \times}$$

$$\boxed{\text{Random Seed} \times \text{Ranking Scheme}}$$

The resulting framework will support systematic, reproducible, statistically validated, and cross-dataset evaluation of machine learning configurations for intrusion detection.

28 Conclusion

This Engineering and Master's project proposes a substantial extension for an existing systematic evaluation study on machine learning-based intrusion detection.

The project will preserve the strengths for the existing experimental protocol, particularly the strict separation between training and test data, while addressing important scientific limitations through repeated experiments, statistical validation, additional datasets, modern machine learning models, class-level multiclass analysis, and sensitivity analysis for multi-criteria ranking.

The final objective is to identify not only high-performing configurations, but configurations whose performance and ranking remain stable across datasets, random seeds, evaluation criteria, and experimental assumptions.

Contact

Students interested in this project are invited to contact:

Dr. Mohammed Bekkouche
École Supérieure en Informatique – Sidi Bel Abbès (ESI-SBA)
LabRI-SBA Laboratory
Email: m.bekkouche@esi-sba.dz